

A Separable Response Covariance for the Covariance-Moderation DGP

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Purpose. The project is converging on a single covariance-based data-generating process (DGP), with an AR(1) within-factor kernel. This note sets out one further choice, whether the correlation among the three response factors across visits should be *separable*, and states the case for and against.

1. The proposal

Each participant's response components TV , PB and BR are observed at visits $t = 1, \dots, n$ at cumulative weeks w_t . A separable structure writes their correlation as a product of a factor part and a time part:

$$\text{Cor}(X_{c,t}, X_{c',s}) = K_{cc'} A_{ts}, \quad A_{ts} = \rho^{|w_t - w_s|}, \quad K = \begin{pmatrix} 1 & c_1 & c_1 \\ c_1 & 1 & c_1 \\ c_1 & c_1 & 1 \end{pmatrix}.$$

With variables ordered factor by factor, as `buildSigma()` already orders them, the response block is the Kronecker product $M = K \otimes A$. The biomarker coupling vector, the baseline term and the variances are unchanged.

Current structure, for comparison. Architecture B (`covar`) uses the same within-factor kernel and same-visit cross-factor correlation c_1 , but sets the cross-factor correlation between *different* visits to $c_\times \rho^{|w_t - w_s|}$ with a separate parameter $c_\times = 0.1 < c_1 = 0.2$. The cross-factor correlation therefore halves between lag 0 and the first positive lag.

What separability asserts. Cross-factor correlation decays at the same rate as within-factor correlation, and is c_1 times it at every lag. Generatively, the three factors are fixed linear mixtures of three independent AR(1) processes that share one persistence ρ .

2. Advantages

1. **The matrix is always valid.** The eigenvalues of $K \otimes A$ are products of those of K and A . K is positive definite for $-0.5 < c_1 < 1$, and the continuous-time AR(1) kernel is positive definite for any visit schedule. No schedule can break the response block. The current structure is not positive definite at half-week spacing, a failure caused by the drop from c_1 to $c_\times$ at the first lag.
2. **No implementation ambiguity.** There is no per-pair cross-factor decay rate, so the loop-order dependence of that rate in the current `buildSigma()` cannot arise.
3. **An exact, factorized ceiling on c_{bm} .** Because $(K \otimes A)^{-1} = K^{-1} \otimes A^{-1}$ and the biomarker couples only to BR , the largest testable moderation parameter is

$$c_{bm}^* = ([K^{-1}]_{33})^{-1/2} (u^\top A^{-1} u)^{-1/2}, \quad u^\top A^{-1} u = u_1^2 + \sum_{t \geq 2} \frac{(u_t - \rho^{d_t} u_{t-1})^2}{1 - \rho^{2d_t}},$$

where u is the coupling pattern and d_t the gap before visit t . The factor term equals $\sqrt{1 - R^2}$ for the regression of BR on TV and PB at one visit ($R^2 = 0.067$ at $c_1 = 0.2$). The schedule term is the switch-cost expression that the ceiling analysis (`docs/34-cbm-feasible-range.md`, Section 5.5) currently offers only as an approximation; under separability it is exact. The ceiling can be computed for every simulation cell before any data are drawn.

4. **A standard structure.** Kronecker-product covariances are a conventional model for multivariate repeated measures, in which several outcomes are recorded at the same visits.
5. **Fewer parameters.** The cross-factor structure needs c_1 only.

3. Disadvantages

1. **Numbers change.** Lagged cross-factor correlations become $c_1\rho^{\text{gap}}$ instead of $c_\times\rho^{\text{gap}}$, which doubles them at production values. Simulated data will not reproduce the current `covar` exactly. The paper 01 Section 3 results must be re-run in any case, because they predate the carryover correction.
2. **One persistence for all factors.** The three factors must share ρ . Production already sets $\rho = 0.7$ for all three, but factor-specific persistence, such as placebo belief decaying faster than natural history, cannot be represented. The principled extension is a sum of separable terms, $M = \sum_j K_j \otimes A_j$, which keeps the validity guarantee but loses the closed-form ceiling.
3. **No lead-lag structure.** Separability excludes one factor predicting another at a later visit more strongly than the reverse. Neither current implementation represents this either.
4. **Less flexibility in cross-lag decay.** Cross-factor correlation cannot decay faster or slower than within-factor correlation. The current structure does impose a faster decay, but its halving at the first lag has no stated rationale in the project documentation.

4. Effect on the largest testable c_{bm}

Ceilings without carryover, at $\rho = 0.7$ and $c_1 = 0.2$ (current structure also uses $c_\times = 0.1$):

Path	Current <code>covar</code>	Separable AR(1)
CO, path A	0.619	0.616
Hybrid, path A	0.456	0.480
Hybrid, path C	0.465	0.491
OL+BDC, path A	0.452	0.474
Multi-cycle, 16 weekly visits	0.262	0.286
Dense, 0.5-week spacing	not positive definite	0.509

The closed form agrees with the full-matrix calculation to 10^{-16} on all six schedules. The change is small in these designs and mostly upward; the principal practical gains are validity and the exact ceiling, not a wider range.

5. Recommendation

Adopt the separable structure for the single covariance-moderation DGP. Its assumption, one clock for all factors with cross-factor correlation proportional to within-factor correlation, is simpler and more defensible than the current lag-0 discontinuity, and it removes the dense-sampling failure and the loop-order ambiguity. Implementation requires replacing the cross-factor loops in `buildSigma()` with `kronecker(K, A)`, retiring `c.cfct`, and adding the closed-form ceiling as a pre-simulation check.

Evidence status. The ceilings and the agreement of the closed form were computed for this note (verified). The positive-definiteness guarantee and the factorization follow from standard Kronecker-product identities. The claim that the current $c_\times < c_1$ choice lacks a documented rationale rests on the documents reviewed, not an exhaustive search. The structure has not yet been implemented or used in a simulation.